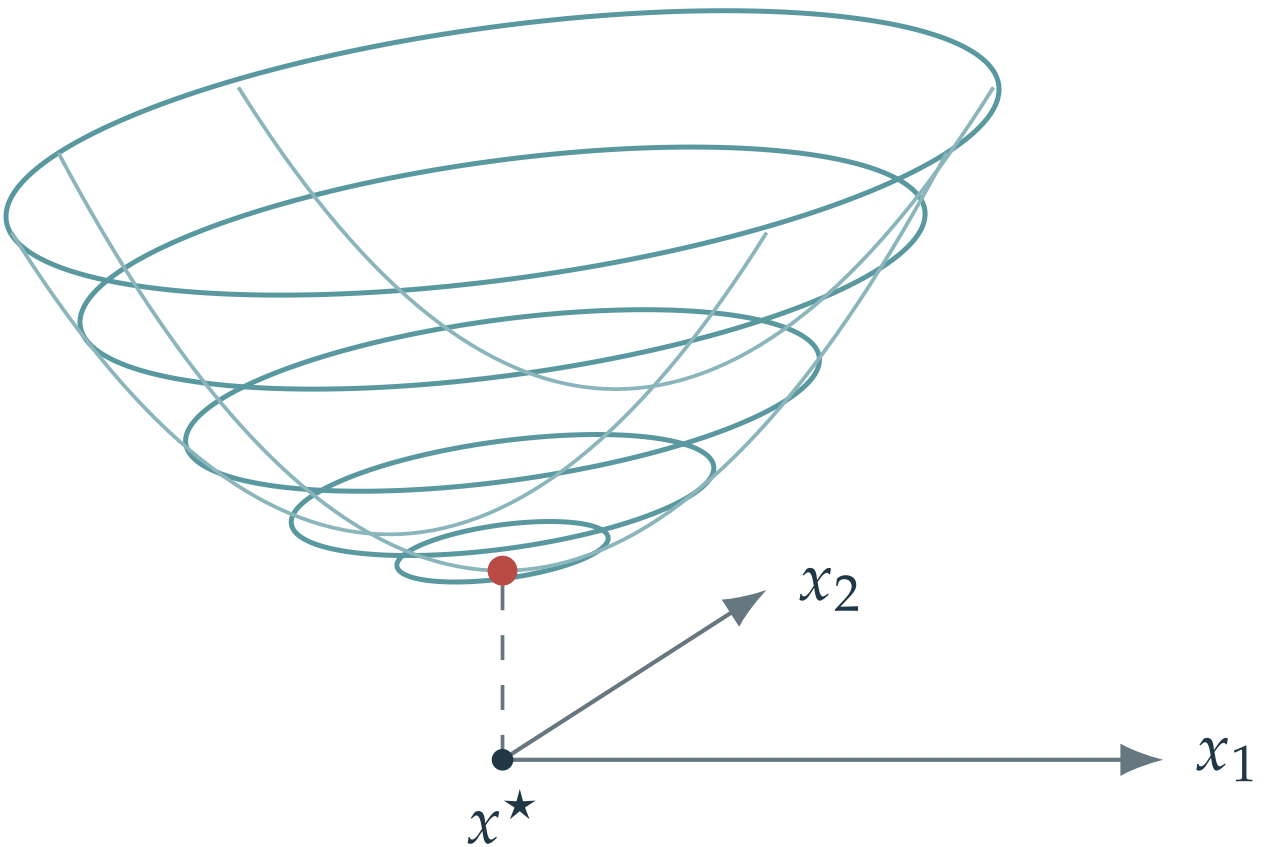
$$f(x) = \tfrac{1}{2}\|Ax - y\|^2$$



$\ker(A) = \{0\}$: unique minimizer